

PROJECTE HOSPITAL DE BLANES

DOCUMENT FINAL D'INSTAL·LACIÓ



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REQUISITS TÈCNICS

Objectiu principal: L'objectiu del disseny és garantir el funcionament ininterromput (24x7) dels serveis assistencials de l'hospital (accés a historials clínics i gestió de visites) i assegurar la protecció de les dades de caràcter confidencial.

Arquitectura de dos nodes (Actiu-Passiu): Es proposa un sistema basat en programari lliure (PostgreSQL i Pgpool-II) estructurat en dues ubicacions físiques diferents:

- **Node 1 (Local/Actiu):** Maquina Principal
- **Node 2 (Cloud/Passiu):** Replica
- **Gestió d'errors (Failover automàtic):** S'utilitza un component intermig (Pgpool-II) que supervisa contínuament l'estat d'ambdós servidors. Si el servidor de l'hospital deixa de funcionar, el sistema redirigeix automàticament tot el trànsit cap al servidor del núvol.

Estratègia de Còpies de Seguretat: Com a complement a l'alta disponibilitat, s'estableix un procés automàtic de còpies de seguretat diàries. Aquestes es guarden de manera local (mantenint les últimes versions) i també s'envien xifrades a un repositori extern.

Aplicacions:

- SO Linux Ubuntu 24.04.3 LTS
- PostgreSQL v14

Llibreries:

- psycopg2
- faker
- random
- datetime
- pyinstaller
- cryptography

INSTAL·LACIONS PRÈVIES

1. Instal·lem el sistema gestor de base de dades al servidor

```
vicifu007@alexplanas:~$ sudo apt install postgresql
[sudo] contrasenya para vicifu007:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
postgresql ya está en su versión más reciente (16+257build1.1).
Los paquetes indicados a continuación se instalaron de forma automática y ya
no son necesarios.
  libllvm19 libwoff1 linux-headers-6.14.0-33-generic
  linux-hwe-6.14-headers-6.14.0-33 linux-hwe-6.14-tools-6.14.0-33
  linux-image-6.14.0-33-generic linux-modules-6.14.0-33-generic
  linux-modules-extra-6.14.0-33-generic linux-tools-6.14.0-33-generic
Utilice «sudo apt autoremove» para eliminarlos.
0 actualizados, 0 nuevos se instalarán, 0 para eliminar y 188 no actualizado
s.
```

2. Utilitzem l'usuari postgres i li creem una contrasenya

```
vicifu007@alexplanas:~$ su postgres
Contraseña:
postgres@alexplanas:/home/vicifu007$
```

CREACIÓ DE LA BASE DE DADES

1. Creem la base de dades del nostre hospital

```
postgres@alexplanas:/home/vicifu007$ CREATE DATABASE hospital_santa_paciencia
```

2. Importem el script de l'estructura (taules, atributs) i l'executem: [enllaç](#)

```
postgres@alexplanas:/home/vicifu007$ CREATE SCHEMA IF NOT EXISTS hospital;
SET search_path TO hospital;

CREATE TABLE IF NOT EXISTS TREBALLADOR (
  id_empleat SERIAL PRIMARY KEY,
  dni VARCHAR(9) NOT NULL UNIQUE,
  nom VARCHAR(50),
  cognom VARCHAR(100),
  telefon VARCHAR(15) NOT NULL,
  direccio VARCHAR(150)
);

CREATE TABLE IF NOT EXISTS MEDIC (
  id_empleat INT PRIMARY KEY,
  especialitat VARCHAR(50),
  estudi VARCHAR(100),
  cv TEXT,
  FOREIGN KEY (id_empleat) REFERENCES TREBALLADOR(id_empleat)
);
```

CREACIÓ DE ROLS

1. Importem ara el script dels rols, usuaris, grant, revoke i l'executem: [enllaç](#)

```
postgres@alexplanas:/home/vicifu007$ -- =====
=====
-- CREACIÓ DE ROLS
-- =====

CREATE ROLE dba;
CREATE ROLE administracio;
CREATE ROLE medic;
CREATE ROLE infermeria;
CREATE ROLE vari;
CREATE ROLE zelador;

-- =====
-- DBA
-- =====

GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA hospital TO dba;
GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA hospital TO dba;
GRANT ALL PRIVILEGES ON ALL FUNCTIONS IN SCHEMA hospital TO dba;

-- =====
-- ADMINISTRACIO
-- =====

GRANT SELECT, INSERT, UPDATE ON hospital.PACIENT TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.RESERVA TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.HABITACIO TO administracio;
GRANT SELECT, INSERT, UPDATE ON hospital.PLANTA TO administracio;
GRANT SELECT ON hospital.TREBALLADOR TO administracio;

-- =====
-- MEDIC
-- =====

GRANT SELECT, INSERT, UPDATE ON hospital.VISITA TO medic;
GRANT SELECT, INSERT, UPDATE ON hospital.RECEPTE TO medic;
GRANT SELECT, INSERT, UPDATE ON hospital.OPERACIO TO medic;

GRANT SELECT ON hospital.PACIENT TO medic;
GRANT SELECT ON hospital.MEDICAMENT TO medic;
```

SSL

1. Instal·lem el openssl

```
vicifu007@alexplanas:~$ sudo apt update
sudo apt install openssl -y
[sudo] contraseña para vicifu007:
Obj:1 http://es.archive.ubuntu.com/ubuntu noble InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Des:5 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2.007 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [355 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Des:8 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3.202 kB]
Des:9 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [740 kB]
Des:10 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1.690 kB]
Des:11 http://es.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [328 kB]
Des:12 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Des:13 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [10,9 kB]
Des:14 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Des:15 http://es.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5.744 B]
Des:16 http://es.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10,6 kB]
```

2. Creem el certificat i la clau privada

[illegible]

3. Donem propietat a l'usuari de la base de dades i que només ell pugui llegir la clau privada
4. I afegim que el certificat pugui ser llegit per tothom

```
vicifu007@alexplanas:~$ sudo chown postgres:postgres /etc/postgresql/ssl/server.key /etc/postgresql/ssl/server.crt
vicifu007@alexplanas:~$ sudo chmod 600 /etc/postgresql/ssl/server.key
vicifu007@alexplanas:~$ sudo chmod 644 /etc/postgresql/ssl/server.crt
```

5. Creem el fitxer per activar el SSL en postgres, modifiquem les següents línies:

```
vicifu007@alexplanas:~$ sudo nano /etc/postgresql/16/main/postgresql.conf
vicifu007@alexplanas:~$ sudo systemctl restart postgresql
```

```
GNU nano 7.2 /etc/postgresql/16/main/postgresql.conf
# -----
# PostgreSQL configuration file
# -----
#
# This file consists of lines of the form:
#
#   name = value
#
# (The "=" is optional.)  Whitespace may be used.  Comments are introduced w
# "#" anywhere on a line.  The complete list of parameter names and allowed
# values can be found in the PostgreSQL documentation.
#
# The commented-out settings shown in this file represent the default values.
# Re-commenting a setting is NOT sufficient to revert it to the default valu
# you need to reload the server.
#
# This file is read on server startup and when the server receives a SIGHUP
# signal.  If you edit the file on a running system, you have to SIGHUP the
# server for the changes to take effect, run "pg_ctl reload", or execute
# "SELECT pg_reload_conf()".  Some parameters, which are marked below,
# require a server shutdown and restart to take effect.
#
# Any parameter can also be given as a command-line option to the server, e.>
# "postgres -c log_connections=on".  Some parameters can be changed at run t>
# with the "SET" SQL command.
#
# Memory units:  B = bytes          Time units:  us = microseconds
#               kB = kilobytes      ms  = milliseconds
#               MB = megabytes      s   = seconds
#               GB = gigabytes      min = minutes
#               TB = terabytes      h   = hours
#                                   d    = days
```

```
# - SSL -

ssl = on
#ssl_ca_file = ''
ssl_cert_file = '/etc/postgresql/ssl/server.crt'
#ssl_crl_file = ''
#ssl_crl_dir = ''
ssl_key_file = '/etc/postgresql/ssl/server.key'
#ssl_ciphers = 'HIGH:MEDIUM:+3DES:!aNULL' # allowed SSL ciphers
#ssl_prefer_server_ciphers = on
#ssl_ecdh_curve = 'prime256v1'
#ssl_min_protocol_version = 'TLSv1.2'
#ssl_max_protocol_version = ''
#ssl_dh_params_file = ''
#ssl_passphrase_command = ''
#ssl_passphrase_command_supports_reload = off
```

6. Reiniciem postgres per guardar la configuració

7. Creem el directori pels certificats i assignem propietat a l'usuari postgres

```
vicifu007@alexplanas:~$ sudo mkdir -p /etc/postgresql/ssl
vicifu007@alexplanas:~$ sudo chown postgres:postgres /etc/postgresql/ssl
vicifu007@alexplanas:~$ sudo chmod 700 /etc/postgresql/ssl
vicifu007@alexplanas:~$
```

8. Creem un fitxer per generar nous certificats cada 365 dies

```
vicifu007@alexplanas:~$ sudo chmod +x /usr/local/bin/renovar-postgres-ssl.sh
```

9. Enganxarem el següent script: [enllaç](#)

```
GNU nano 7.2 /usr/local/bin/renovar-postgres-ssl.sh
#!/bin/bash

DIES=365
RUTA="/etc/postgresql/ssl"
IP_VM="192.168.56.103"

openssl req -x509 -nodes -days $DIES -newkey rsa:4096 \
  -keyout $RUTA/server.key \
  -out $RUTA/server.crt \
  -subj "/CN=$IP_VM"

chown postgres:postgres $RUTA/server.key $RUTA/server.crt
chmod 600 $RUTA/server.key
chmod 644 $RUTA/server.crt

systemctl restart postgresql
```


10. Accedim al crontab per automatitzar els backups

```
vicifu007@alexplanas:~$ sudo crontab -e
```

11. Afegim la següent línia al final:

```
GNU nano 7.2 /tmp/crontab.b9JnTx/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
9 # m h dom mon dow command
0 0 1 1 * /usr/local/bin/renovar-postgres-ssl.sh >> /var/log/renovacio-ssl-p
00 03 * * * /home/el_teu_usuari/scripts/backup_bd.sh
```

12. Verifiquem que el ssl estigui activat:

```
vicifu007@alexplanas:~$ sudo -u postgres psql
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# SHOW ssl;
 ssl
-----
 on
(1 row)
```

DATA MASKING

1. Executem el següent script amb l'usuari postgres per crear les vistes per a vari i zelador: [enllaç](#)

```

1  |--VISTA VARI
2
3  CREATE OR REPLACE VIEW hospital.v_pacient_vari AS
4  SELECT
5  id_pacient,
6  'XXXXX' || RIGHT(dni, 4) AS dni,
7  '*****' || RIGHT(telefon, 3) AS telefon,
8  LEFT(email, 1) || '*****@' || SPLIT_PART(email, '@', 2) AS email,
9  nom,
10 cognom
11 FROM hospital.pacient;
12
13 GRANT SELECT ON hospital.v_pacient_vari TO vari;
14
15 --VISTA ZELADOR
16
17 CREATE OR REPLACE VIEW hospital.v_pacient_zelador AS
18 SELECT
19 id_pacient,
20 'XXXXX' || RIGHT(dni, 4) AS dni,
21 '*****' || RIGHT(telefon, 3) AS telefon,
22 LEFT(email, 1) || '*****@' || SPLIT_PART(email, '@', 2) AS email,
23 nom,
24 cognom
25 FROM hospital.pacient;
26
27 GRANT SELECT ON hospital.v_pacient_zelador TO zelador;
28

```

REPLICACIÓ

1. Creem l'usuari de replicació: [enllaç](#)

```

vicifu007@alexplanas:~$ sudo -u postgres psql -c "CREATE ROLE replica_user RE
PLICATION LOGIN PASSWORD '1234';"
ERROR:  role "replica_user" already exists

```

2. Editem el fitxer de configuració i canviem les següents línies:

```
sudo nano /etc/postgresql/16/main/postgresql.conf
```

```
#-----
# REPLICATION
#-----

# - Sending Servers -

# Set these on the primary and on any standby that will send replication data

max_wal_senders = 10          # max number of walsender processes
                              # (change requires restart)
#max_replication_slots = 10   # max number of replication slots
                              # (change requires restart)
#wal_keep_size = 0            # in megabytes; 0 disables
#max_slot_wal_keep_size = -1  # in megabytes; -1 disables
#wal_sender_timeout = 60s     # in milliseconds; 0 disables
#track_commit_timestamp = off # collect timestamp of transaction commit
                              # (change requires restart)
```

```
#-----
# CONNECTIONS AND AUTHENTICATION
#-----

# - Connection Settings -

listen_addresses = '*'        # what IP address(es) to listen on;
                              # comma-separated list of addresses;
                              # defaults to 'localhost'; use '*' for all
                              # (change requires restart)
```

```
#----->
# WRITE-AHEAD LOG
#----->

# - Settings -

wal_level = replica           # minimal, replica, or logical
```

3. Editem el fitxer d'autenticació:

```
vicifu007@alexplanas:~$ sudo nano /etc/postgresql/16/main/pg_hba.conf
vicifu007@alexplanas:~$ sudo systemctl restart postgresql
```

```
GNU nano 7.2 /etc/postgresql/16/main/pg_hba.conf
#
#-----
```

4. Afegim la següent línia al final:

5. Reiniciem per aplicar canvis

6. Instal·lem postgres a la nova màquina:

```
bd-replica@bd-replica-VirtualBox:~$ sudo apt update
[sudo] contraseña para bd-replica:
Des:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Obj:2 http://archive.ubuntu.com/ubuntu noble InRelease
Des:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1.69
8 kB]
Des:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2.007
kB]
Des:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [267
kB]
Des:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [42
```

```
bd-replica@bd-replica-VirtualBox:~$ sudo apt install postgresql-16 postgresql-cl
ient-16 -y
```

7. Aturem el servei i buidem el directori:

```
bd-replica@bd-replica-VirtualBox:~$ sudo systemctl stop postgresql
[sudo] contraseña para bd-replica:
bd-replica@bd-replica-VirtualBox:~$ sudo rm-rf /var/lib/postgresql/16/main/*
```

8. Sincronitzem la base de dades des de la màquina principal

```
vicifu007@alexplanas:~$ sudo -u postgres pg_basebackup -h 192.168.56.103 -D
/var/lib/postgresql/16/main/ -U replica_user -P -R
Password:
```

9. Tornem a posar en marxa a la replica:

```
bd-replica@bd-replica-VirtualBox:~$ sudo systemctl start postgresql
```

10. Verifiquem que s'ha sincronitzat correctament creant el següent al node 1

```
viciifu007@alexplanas:~$ sudo -u postgres psql -c "CREATE TABLE comprobacio_final (id SERIAL PRIMARY KEY, missatge TEXT);"
sudo -u postgres psql -c "INSERT INTO comprobacio_final (missatge) VALUES ('Replicacio funcionant correctament');"
ERROR: relation "comprobacio_final" already exists
INSERT 0 1
```

11. Mirem des de el node 2 que s'hagi replicat:

```
bd-replica@bd-replica-VirtualBox: ~
bd-replica@bd-replica-VirtualBox:~$ sudo -u postgres psql -c "SELECT * FROM comprobacio_final;"
[sudo] contraseña para bd-replica:
id |          missatge
---+-----
  1 | Replicacio funcionant correctament
(1 row)
```

BACKUPS

1. Creem el fitxer per a la creació de backups:

```
vicifu007@alexplanas:~$  
nano ~/scripts/backup_bd.sh
```

2. Enganxem el següent contingut: [enllaç](#)

```
GNU nano 7.2 /home/vicifu007/scripts/backup_bd.sh  
#!/bin/bash  
# Script de Backup - Hospital Santa Paciencia  
  
# Configuració  
DATE=$(date +%Y%m%d_%H%M%S)  
BACKUP_DIR="/var/backups/postgres"  
DB_NAME="hospital_santa_paciencia" # Assegura't que aquest és el nom de la  
KEEP_DAYS=5  
  
# Crear el directori de destí si no existeix  
sudo mkdir -p $BACKUP_DIR  
sudo chown postgres:postgres $BACKUP_DIR  
  
echo "Iniciant backup de $DB_NAME..."  
  
# Executar el backup  
# Nota: fem servir sudo per executar-ho com l'usuari postgres  
sudo -u postgres pg_dump $DB_NAME > $BACKUP_DIR/backup_$DATE.sql  
  
# Rotació: Esborrar fitxers de més de 5 dies (Requeriment 3)  
sudo find $BACKUP_DIR -type f -mtime +$KEEP_DAYS -name "*.sql" -delete  
  
echo "Backup finalitzat a $BACKUP_DIR/backup_$DATE.sql"
```

3. Comprovem que s'executen els backups i es guarden al directori:

```
vicifu007@alexplanas:~$ chmod +x ~/scripts/backup_bd.sh
vicifu007@alexplanas:~$ sudo ~/scripts/backup_bd.sh
Iniciant backup de hospital_santa_paciencia...
Backup finalitzat a /var/backups/postgres/backup_20260525_173359.sql
vicifu007@alexplanas:~$ ls -lh /var/backups/postgres
total 17M
-rw-r--r-- 1 root root 8,2M may 25 17:33 backup_20260525_173339.sql
-rw-r--r-- 1 root root 8,2M may 25 17:34 backup_20260525_173359.sql
vicifu007@alexplanas:~$ |
```

4. Editem el crontab -e i afegim la següent línia per automatitzar-ho a las 3 a.m.:


```
GNU nano 7.2 /tmp/crontab.EoyxXm/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
00 03 * * * /home/vicifu007/scripts/backup_bd.sh
```